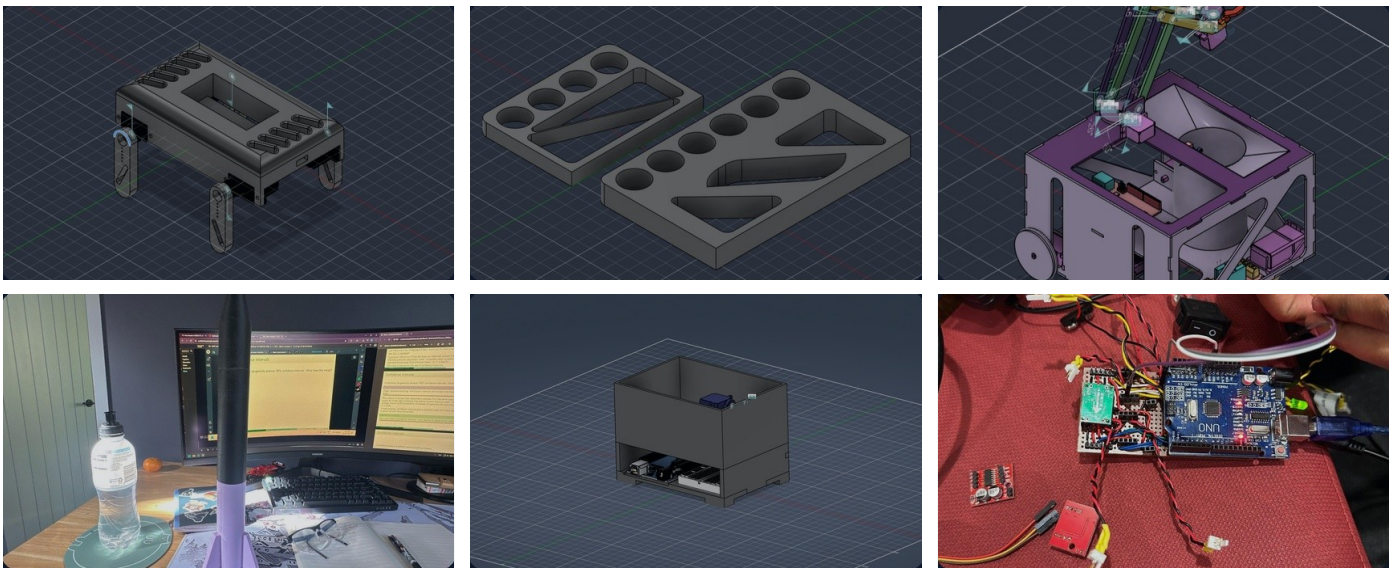


MECHANICAL DESIGN / EMBEDDED SYSTEMS / PROTOTYPING

Engineering **Project Portfolio**

Matthias Paterson-Hawes

Mechanical Engineering Student

CAD & MECHANICAL DESIGN	Fusion 360, SolidWorks, assemblies and tolerance design
EMBEDDED SYSTEMS	Arduino, ESP32, sensors, servos and electromechanical integration
PROTOTYPING & PROGRAMMING	FDM printing, C/C++, Python, MATLAB, testing and design iteration

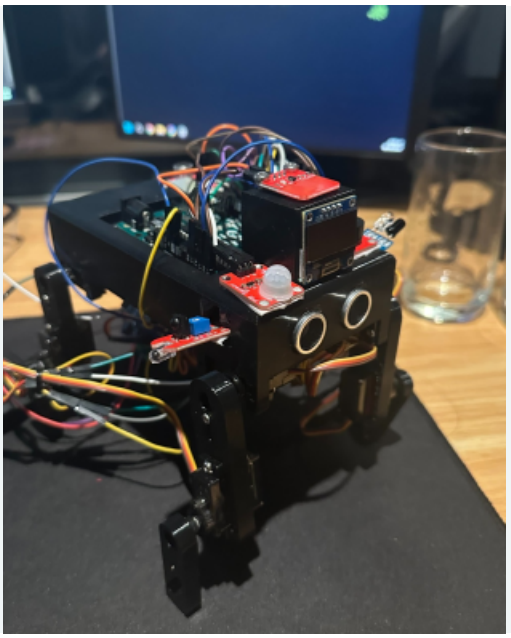
Selected work demonstrating the full engineering process: defining a need, developing a concept, designing interfaces, manufacturing prototypes, integrating electronics, testing performance and refining the result.

GITHUB **github.com/MatthiasPH**

PROJECT 01

Quadruped Robot - Version 2

ESP32-C3 control | 8-servo 2-DOF legs | Sensor integration



Assembled second-version quadruped prototype.

PROJECT OVERVIEW

I redesigned my original quadruped into a more capable second version using eight MG90S servos, giving each leg independent hip and knee movement. An ESP32-C3 coordinates the gait, while onboard sensing provides a platform for obstacle detection and autonomous behaviour.

USEFULNESS

Version 2 turns the first walking prototype into a more realistic robotics platform. Two joints per leg provide greater control of foot placement and body motion, while the modular sensor layout makes it easier to test new behaviours without redesigning the complete chassis.

WHAT I LEARNED

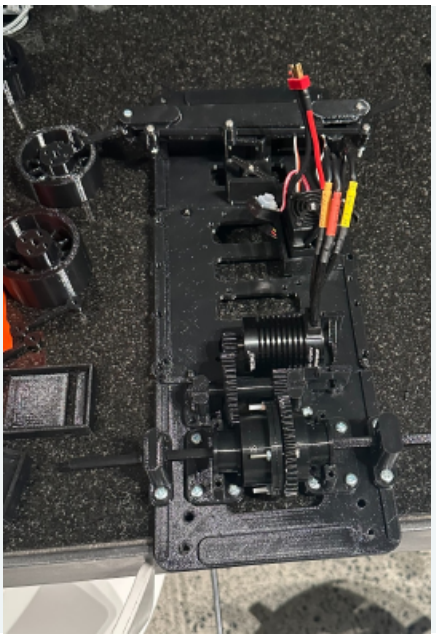
- Expanded from a simpler leg mechanism to two controlled joints per leg and developed coordinated multi-servo gait sequences.
- Calibrated individual servo centres and travel limits so mechanical motion matched commanded angles more reliably.
- Worked through power-distribution and transient-load issues caused by multiple servos moving at the same time.
- Improved component packaging, cable routing and sensor placement while keeping the chassis accessible for debugging.

SKILLS ESP32-C3 / Embedded C/C++ / 8-servo control / Gait development / Sensor integration / CAD/FDM / System debugging

PROJECT 02

Custom RC Car

3D-printed chassis | Custom geartrain | Drivetrain prototyping



Printed chassis and drivetrain assembly.

PROJECT OVERVIEW

I designed and built a custom RC car around a 3D-printed chassis and electric drivetrain. The project involved CAD modelling, drivetrain layout, gearing calculations, printed gearbox and motor mounts, shaft support, assembly and repeated testing to refine the mechanical system.

USEFULNESS

The car gave me a compact platform for applying mechanical design to a rotating drivetrain where alignment, stiffness and tolerances directly affect performance. The modular printed chassis also allows parts to be repositioned, replaced and iterated quickly during development.

WHAT I LEARNED

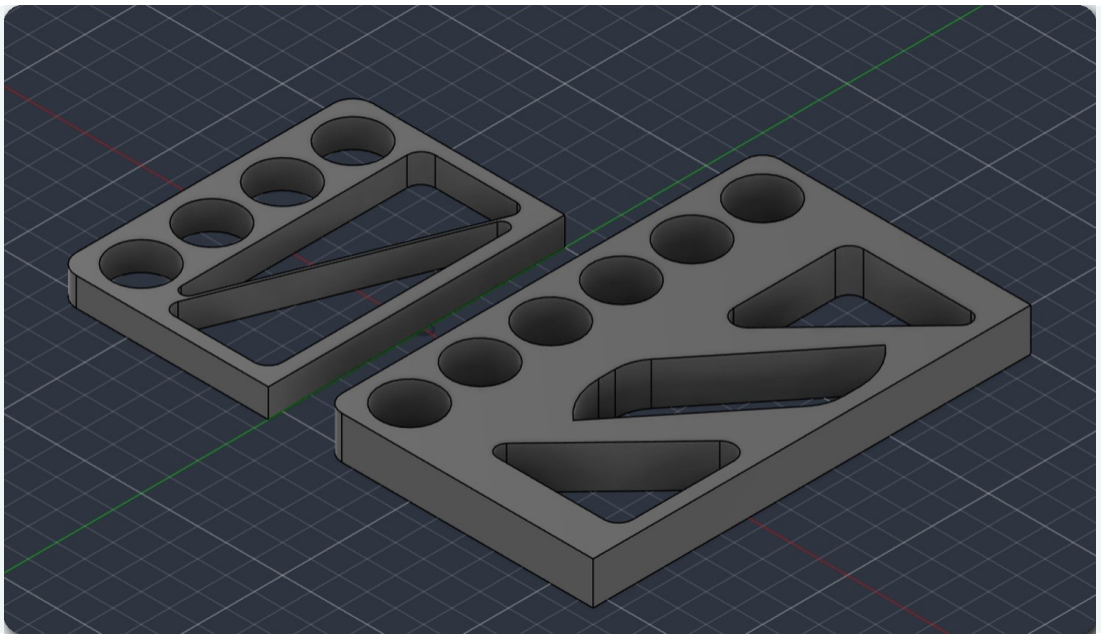
- Applied gear-ratio and centre-distance calculations to convert the required reduction into a physical geartrain and package it inside the chassis.
- Developed practical backlash, shaft-support and bearing arrangements so printed components could transmit torque without binding.
- Saw how print orientation, dimensional error and local stiffness affect gear mesh, alignment and drivetrain reliability.
- Used assembly and test failures to identify weak interfaces, revise mounts and improve serviceability for later iterations.

SKILLS CAD modelling / Geartrain design / Gearing calculations / 3D printing / Drivetrain packaging / Assembly / Troubleshooting

PROJECT 03

Modular Dowel Storage Mounts

20 mm dowel interfaces | Parametric CAD | Tolerance development



CAD models of alternative dowel mount geometries.

PROJECT OVERVIEW

I created a family of 3D-printed mounts that locate 20 mm wooden dowels and turn them into a modular storage structure. Multiple geometries were developed so the same dowel size could be used in different support and bracing arrangements.

USEFULNESS

The design is inexpensive, repairable and easily adapted to different spaces. Individual mounts can be resized or reprinted instead of replacing the complete storage system, making the concept suitable for rapid, custom organisation.

WHAT I LEARNED

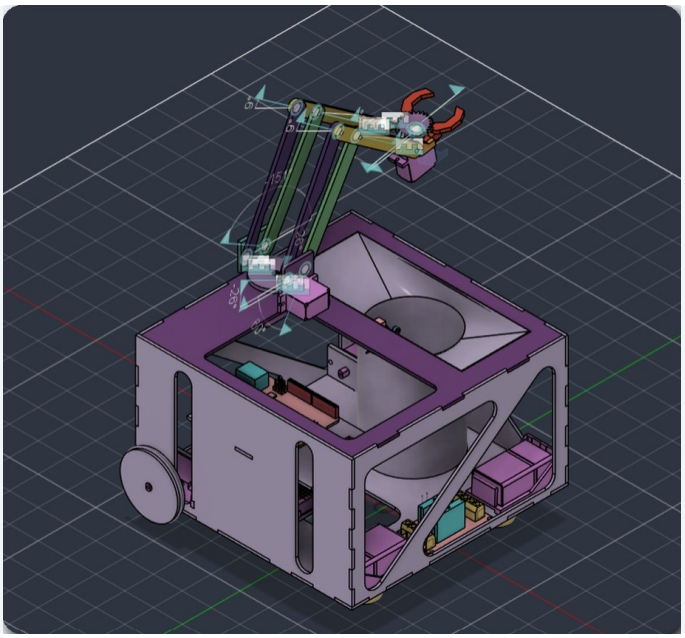
- Developed practical clearances for slip fits and interference fits rather than relying only on nominal CAD dimensions.
- Saw how print orientation, first-layer expansion, shrinkage and machine calibration affect the final interface.
- Used small test pieces and repeated iterations to converge on reliable tolerances while reducing wasted material.
- Improved load-path design with wall thickness, fillets, bracing and material placed where it contributes most to stiffness.

SKILLS Tolerance design / Parametric CAD / Design for FDM / Fit testing / Structural bracing / Iterative prototyping

PROJECT 04

Warman Project Concept Robot

University design concept | Integrated mechanism | Design for assembly



Digital concept assembly with mobile chassis, hopper and gripper arm.

PROJECT OVERVIEW

For a university Warman design project, I developed a detailed CAD concept combining a mobile chassis, collection hopper and multi-link arm with a gripper. The concept was not selected for the final build, but completing the assembly required substantial mechanism and interface design.

USEFULNESS

A complete digital assembly allows a team to evaluate packaging, reach, interference and assembly order before committing time and material to manufacture. It also exposes weak interfaces that may not be visible in isolated part models.

WHAT I LEARNED

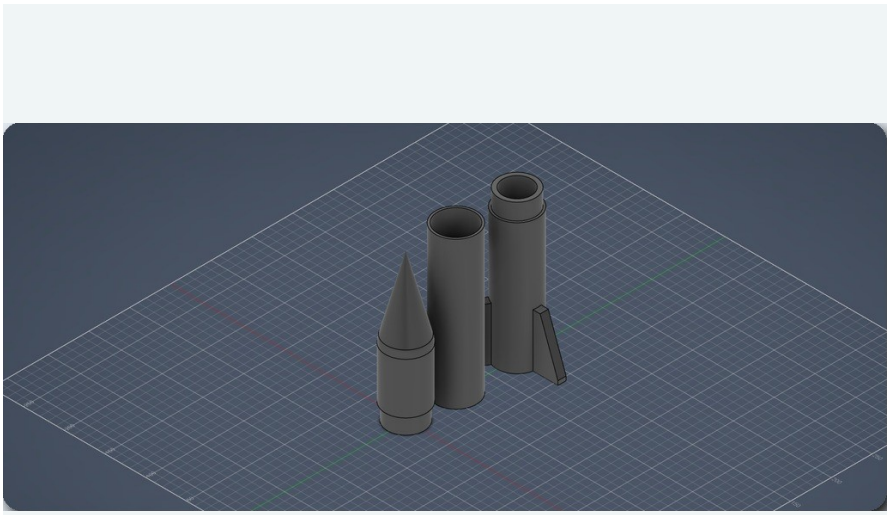
- Used top-down assembly thinking to keep several moving subsystems consistent with the overall machine envelope.
- Learned that a visually correct CAD model is not enough; joints need realistic fasteners, clearances, support and access.
- Considered how each part would actually be manufactured, aligned, joined and replaced during a real build.
- Improved my ability to convert a concept into a coherent system while identifying collision and integration risks early.

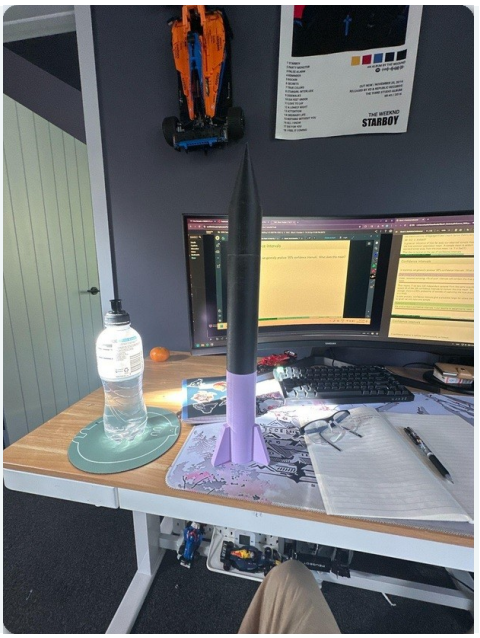
SKILLS SolidWorks assemblies / Mechanism layout / Design for assembly / Interface design / Packaging / Concept evaluation

PROJECT 05

Modular Two-Stage Rocket

Fusion 360 | 3D-printed prototype | Modular subsystem architecture





Modular CAD components and the assembled printed prototype.

PROJECT OVERVIEW

This was my first iteration of a small modular two-stage rocket. The first stage uses fins for passive stability, while the second-stage concept was intended to use gyroscope-controlled thrust vectoring. I modelled the components in Fusion 360 and assembled a full-size printed prototype.

USEFULNESS

The modular architecture allows sections to be removed, inspected and redesigned independently. A physical prototype verifies fit, scale and packaging before progressing to more complex sensing and control development.

WHAT I LEARNED

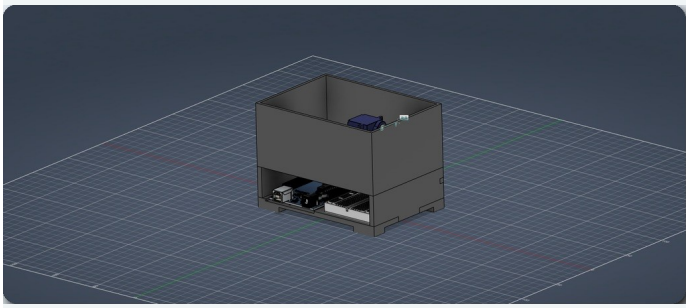
- Designed repeatable mating interfaces and adjusted clearances until the printed sections assembled accurately.
- Developed an understanding of tolerance stack-up across several cylindrical and nested components.
- Considered stability, internal packaging and the trade-off between ambitious functionality and manufacturability.
- Used the physical build to identify geometry and assembly issues that were not obvious from the CAD model alone.

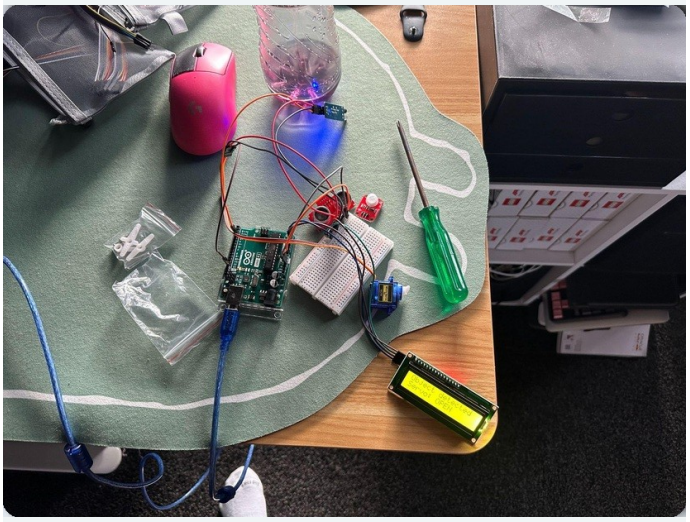
SKILLS Fusion 360 / Tolerance stack-up / Modular design / 3D printing / Physical validation / Design iteration

PROJECT 06

Automatic Rubbish Bin

Arduino | IR proximity sensing | Servo-actuated lid





Enclosure CAD model and electronics test setup.

PROJECT OVERVIEW

I built a touch-free rubbish bin using an Arduino, an IR proximity sensor, a display and an SG90 servo. When a user is detected, the servo opens the lid and the control logic keeps it open for approximately two seconds before closing.

USEFULNESS

The bin provides hygienic hands-free operation and solved a real problem in my workspace by containing accumulated 3D-printing waste. It combines sensing, control and mechanical actuation in a compact everyday product.

WHAT I LEARNED

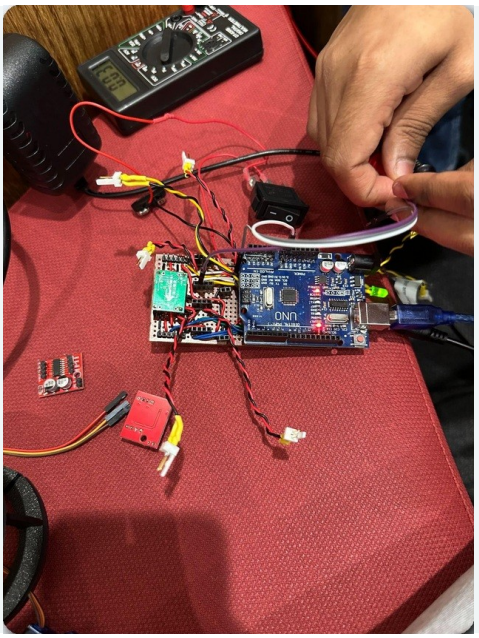
- Integrated a sensor, display and actuator into a single event-driven control sequence.
- Implemented timing behaviour so the lid remained open without blocking the rest of the program.
- Designed the enclosure and mechanism around electronics access, servo travel and reliable lid movement.
- Tested sensor placement and trigger behaviour to reduce missed detections and unintended activation.

SKILLS Arduino / IR sensing / Servo actuation / Control logic / Enclosure design / Prototype testing

PROJECT 07

Modular Electronics Control Board

Hand-assembled integration | Motor drivers | Regulated power distribution



Hand-assembled control board during integration and testing.

PROJECT OVERVIEW

I designed and soldered a compact control board integrating an Arduino Uno, two motor-driver modules, a 5 V buck converter and removable connectors. The board consolidated a larger wiring arrangement while keeping major components replaceable.

USEFULNESS

Centralising the electronics reduces loose wiring, improves repeatability and makes troubleshooting easier. Maintaining modular connections also means a failed driver or controller can be replaced without rebuilding the complete electrical system.

WHAT I LEARNED

- Planned signal routing and power distribution around modules with different voltage and current requirements.
- Improved soldering, continuity checking and systematic fault-finding with a multimeter.
- Positioned connectors for assembly access, serviceability and clearer cable management.
- Learned how modular design and failure isolation improve reliability during prototype development.

SKILLS Circuit integration / Soldering / Motor drivers / Buck conversion / Power distribution / Troubleshooting